News Shocks in Prediction Markets

Real results, 2026-10-13

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The question driving this thesis

Do news shocks about a company produce a measurable, predictable price reaction in that company's Polymarket contracts?

Everything that follows answers this question directly.

Context

- Data: 6,928 resolved contracts across 86 companies (Apr 2024-Aug 2026), matched against 546,657 GDELT articles
- No existing tool answers the question above directly, so this project builds two new methods
- GDELT only timestamps articles to the day, so every result below measures the 24-hour price reaction, the only window this feed supports

Method: the mechanics

Matching picks the right article for a contract, combining text similarity and trading-volume spikes:

match_quality = w1 · logit(cosine similarity) + w2 · z(volume spike), then sigmoid

Consistency forces a company's related contracts (above $150, $160, $170) to react in the same direction, weighted by volume, via isotonic regression (PAVA).

Method: why these, and why they should work

Both methods are new combinations of established tools, not inventions from scratch.

Combining independent signals in log-odds space is standard practice (the same math behind naive Bayes classifiers). Volume spikes as a marker of real news arriving is a standard idea in market microstructure research, going back to classic event-study methodology. Isotonic regression is a long-established tool for order-restricted inference (Barlow, Bartholomew, Bremner & Brunk, 1972), and the logic that related contracts should price consistently is the same logic behind no-arbitrage pricing.

What's new: applying both combinations to this specific problem, matching news to prediction-market contracts and enforcing no-arbitrage-style consistency across a strike ladder, which we are not aware of being done before.

Result 1: contracts do react to news

Method	Spearman ρ vs. actual price move	n
Matching score (text + volume)	**0.509**	14,717
Text similarity alone	0.233	14,717

This is half the answer to our question: a company's news and its market's trading behavior are measurably linked. Every later result depends on this step, it is how we built a trustworthy dataset of article-reaction pairs at all.

Result 2: the consistency correction doesn't help, so we don't use it

Category	R^2, raw reaction	R^2, after correction
Price-target	0.084	0.073
Valuation	0.746	0.363

We expected this to help: related contracts should price consistently, by no-arbitrage logic. It likely hurts because the correction smooths away real, contract-specific variation the prediction models depend on. We predict from the raw reaction, not the corrected one.

Result 3: but the reaction isn't predictable in advance

Category	Best model	R^2	n (test)
All (pooled)	Linear	−0.010	5,151
Corporate event	Linear	−0.021	3,157
Price-target	LSTM	**+0.010**	598
Revenue-target	TCN	−0.016	9
Valuation	Linear only	−0.255	small

This is the other half of the answer. If Polymarket prices already reflect public news quickly, this is exactly the pattern we'd expect: a real reaction (Result 1), but no way to forecast its size from the article alone.

Result 4: one of four pre-registered predictions holds up

Prediction	Tests whether…	Result
Richer embeddings beat simpler ones	text representation matters	Not supported
Best model differs by category	architecture choice matters	Not supported
Quantitative articles beat high-attention ones	article type matters	Supported (n=28)
Best ladder category beats corporate events	ladder structure adds signal	Not supported

Tested with bootstrap resampling clustered by real-world event, so duplicate coverage of one story isn't counted as independent evidence.

So, does Polymarket respond to news shocks?

Yes, but only partly. There is a real, strong relationship between an article and the price move it precedes (Result 1): when we can identify the right article, the market has reacted.

That reaction is not predictable in advance. Every model found essentially no exploitable signal (Results 2-4). The market absorbs news efficiently enough that we cannot forecast the size or direction of the reaction ahead of time, with the tools and data available here.

Scoping decisions

- **market_cap_ladder** (4 markets): dropped, too small to mean anything
- **valuation_ladder** (110 markets): kept, flagged as below our 200-market floor, everything else about it clears comfortably
- **revenue_ladder**: given its own data split so it could be trained at all, rather than excluded

Limitations

- GDELT is free but noisy, and day-precision is a structural limit of this feed, not a gap to close
- `resolved_at` and `parent_event_id` are each about 40% backfilled from Polymarket's API; fallbacks cover the rest
- The revenue-target result rests on 19 training examples: real, but small

Recommendation

- Lead the thesis with the matching result (Result 1): it is the clean, strong, genuine contribution
- Present the consistency correction and the model comparison as honest negative results, not failures
- Keep the 200-market floor as the rule, with `valuation_ladder` as a documented, explicit exception
- A licensed newswire is the single highest-leverage next step if budget allows; everything else here is close to final